EVERY TEMPLATE ENCODES A FINITELY BRANCHING OMEGA-TREE, UNIQUELY UP TO LOCAL ISOMORPHISM, WITH BRANCHES CONTROLLED BY BRANCHES OF THE TEMPLATE

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Allowed standard prerequisites (editorial).

- The notion of a template and of a tree encoded by it, with the levels P_n and the attached positive integers. Set up in the definitions preceding the counted statement (the notation P_n, L_n and the partition correspondence used in the proof). Fixes the data that the four clauses quantify over and the meaning of local embeddability used in clause (ii).
- The Fraenkel-Mostowski construction with finite supports used earlier in the paper. Used inside the counted proof of clause (iv) at lines 620-625. Supplies the permutation model in which a tree encoded by a template without an eventually singleton branch has no infinite branch.

Lemma 0.1. *(i) For any template $(T^*, \prec)$ there is a finitely branching ω-tree $(T, <)$ encoded by $(T^*, \prec)$.*

(ii) Any two trees encoded by $(T^*, \prec)$ are locally isomorphic.

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(iii) If $(T^*, \prec)$ has an eventually singleton ω-branch, then $(T, <)$ has an infinite branch.

(iv) If $(T^*, \prec)$ is a template having no eventually singleton branch, then in a suitable Fraenkel–Mostowski model, there is a tree encoded by $(T^*, \prec)$ having no infinite branch.

Proof. (i) Let $(T^*, \prec)$ be a template with nth level P_n and positive integers attached as in the definition. We build $(T, <)$ level by level, using induction. There will be a bijection between P_n and a partition π_n of L_n. Choose any root for T, and let π_0 be the trivial singleton partition of L_0 which just contains the root, and P_0 corresponds to π_0.

Assuming that L_n has been chosen, as has a partition π_n of L_n which corresponds to P_n under some bijection, for $X \in P_n$ let $Y_1, \ldots, Y_k$ be the members of P_{n+1} above X under $\prec$, with positive integers $n_1, \ldots, n_k$ attached. Under the bijection given by the induction hypothesis, X may be viewed as a subset of L_n. For each $x \in X$, and i between 1 and k, choose n_i new points which together (as i varies) will form $succ(x)$ in T. Doing this for all x determines L_{n+1}. The $(n+1)$th level P_{n+1} of T^* is now identified with a partition π_{n+1} of L_{n+1} by placing y_1 and y_2 in the same piece of π_{n+1} provided that for some $X \in \pi_n$, $x_1, x_2 \in X$, and i between 1 and k, y_1 and y_2 are adjoined corresponding to that value of i.

(ii) We observe that we have the freedom to choose the new points adjoined with or without an ordering. Thus if we are (secretly) thinking of them as ordered, then the whole of the tree can be well-ordered, and has cardinality $\aleph_0$. If we make no attempt to order them, then there is no reason why this should be true. So we do not expect the two possibilities to give rise to isomorphic trees. It is clear however that they will be locally isomorphic, since their behaviour is identical up to any finite stage.

(iii) As remarked above, a branch of the template corresponds to a balanced subtree of T, and saying that this is eventually singleton just says that this subtree is eventually just a single branch.

(iv) What we really want to say is that $(T, <)$ has an infinite branch if and only if its template has an eventually singleton branch, but for the reasons explained this is too strong. Suppose then that $(T^*, \prec)$ is a template and that it encodes $(T, <)$. We form a Fraenkel–Mostowski model in which U is indexed by the members of T, and we let G be the group of automorphisms of U induced by tree automorphisms of $(T, <)$ which fix each member of T^*, and we let $\mathfrak{N}$ be the FM-model defined using finite supports. One sees that G acts on each subtree obtained as above from an ω-branch of T^*. This is a balanced tree which has infinitely many non-singleton entries, and so as for the simplest construction given from just one balanced tree, has no infinite branch in $\mathfrak{N}$. Since any infinite branch of T would have to arise in this way, there cannot be any at all. □